



Durval Polónia

Software Engineer

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Experience

Junior Software Engineer, Critical Software, Lisbon Feb 2026 - Present

- Maintained and expanded the API bridging BMW's internal product data with seller's user-facing applications for all markets.
- **Improved CI/CD pipelines** and developed backend services in **Java** using the **Quarkus** framework, with test coverage built on **JUnit**, **Cucumber**, and **Testcontainers**.
- Redesigned and automated application infrastructure across multiple **AWS services** using **CloudFormation**.
- Used state-of-the-art **Agentic AI** to boost productivity and explored novel agentic-based development workflows.

Research Fellow, University of Aveiro, Aveiro Apr 2025 – Jan 2026

- Developed a complete end-to-end pipeline to train syscall-based **anomaly detection ML models** for serverless environments.
- Built an eBPF-based system call interception system to support data acquisition on low-level function behavior.
- Created a dataset of function behavior supported by an in-depth analysis of the vulnerabilities associated with **cloud environments**.

Apprentice Backend Developer, Sky Portugal, Aveiro Jul 2024 – Aug 2024

- Boosted inter-team cooperation by automating technical tasks for product team needs.
- Developed **Kotlin** code for Peacock's **large-scale environment** with thousands of active daily users.
- Designed and documented the complete project lifecycle following **SCRUM** methodology.

QA & Test Automation Summer Intern, adidas, Porto Jul 2023 – Aug 2023

- Used **AWS** to integrate a new data analysis module within the existing cloud infrastructure.
- Explored different **containerization** approaches for module deployment.
- Created data-centric views into the value provided by the test automation team.
- Used **Kanban** as the development methodology.
- Co-presented the scope of the internship to hundreds in adidas's Porto community.

Academic

MSc. Software Engineering, University of Aveiro Sept 2023 – Dec 2025

- **Thesis:** Leveraging system call information to provide function-specific protection in serverless environments.

Erasmus, University of Padova Feb 2024 – Jul 2024

BSc. Computer and Informatics Engineering, University of Aveiro Sept 2020 – Jul 2023

Publications

FADO: Detecting Anomalies in Serverless Functions via eBPF System Call Tracing 2026

IEEE Network Operations and Management Symposium (IEEE NOMS)

Durval P., Escalreira P., Cunha V., Gomes D., Aguiar R.

DOI: Not available yet

Projects

Nvideo — Your local movie database.

2026 (In Progress)

- Built a containerized full-stack movie-catalogue app with a **Django REST** + **PostgreSQL** backend, and automated ingestion from the TMDB API.
- Architected an event-driven reporting subsystem: a standalone **Flask** + **MongoDB** microservice kept in sync from **PostgreSQL** asynchronously through **Celery** tasks over **RabbitMQ**.
- Built and orchestrated a modular **Docker Compose** stack, complemented by fast, mock-based **Python** test suites for reliable testing.
- *Python · Django/DRF · Flask · PostgreSQL · MongoDB · Celery · RabbitMQ · Docker Compose · nginx*

Find it here: github.com/Durval8/nvideo

Technologies & Skills

Programming Languages: Python, Java, C++, Kotlin, SQL.

Technologies: Docker, Git, Linux, Spring-Boot, Quarkus, Kafka, CloudFormation, MongoDB, PostgreSQL, Celery, RabbitMQ, Django.

Areas of Knowledge: Agentic AI, Cloud Computing, DevOps, OOP, Algorithms & Data Structures, Databases, Agile, Machine Learning, Telecommunications & Networking.

Languages: English, Portuguese, Spanish